Class Activity

Part 1

Question: Why is storing the same data multiple times in a table a bad idea?

- It causes data redundancy, which wastes storage space and makes databases harder to manage. It also leads to data inconsistency.

Part 2

1. CSE in DeptName
   Block A in DeptOffice

2. Update Anomaly: When same data is stored in multiple places and must be updated everywhere.
   * If CSE office changes, you must update s1 and s2 rows.

2. Insertion Anomaly: When you cannot insert data without having other data.

- Deletion Anomaly: When deleting a record removes useful information unintentionally.

Part 3

First Normal Form (1NF) rules:

- Each cell must have single (atomic) value
- No repeating groups
- Each row is unique.

The table satisfies these conditions ∴ it is in 1NF.

Table 1: Student

Student ID	Student Name	DeptName
s1	Sonam	CSE
s2	Karma	CSE
s3	Pema	ECE

Table 2: Department.

DeptName	DeptOffice
CSE	Block A
ECE	Block B

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Discussion Questions

1. For Department informations (DeptName, Deptoffice) was ~~rep~~ repeating for multiple students, which causes data redundancy.

2. If solved data anomalies and redundancy. Repeating datas (CSE and Block A) is stored once and don't need to update multiple students rows. We can also add a new department and deleting a student do not remove department data.

Part 4.

Given:

R (A, B, C)

A → B

R1 (A, B)
R2 (A, C)

Soln: A → B (B depends on A as if we know A, we ~~can~~ can determine B)

A determines B

R1 contains A, & B

R2 " A, C

R1 (A, B) ∩ R2 (A, C) = A (common attribute)

A decomposition is lossless if " " is a key in atleast one relation.

In R1 (A, B), A → B

∴ A determines all attributes of R1

∴ A = key of R1

∴ lossless join condition is satisfied.

Check Dependency Preservation:

A → B (B depend on A)

In R1 dependency is preserved.

∴ The decomposition is correct since it is lossless and dependency is preserved.

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Part 5

* 1NF removes repeating groups and non-atomic values (multi-valued attributes)
* 2NF removes partial dependency
* 3NF " transitive "

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